Multimodal AI

Multimodal AI is AI that can understand and work with more than one type of input, such as text, images, audio, or video.

These different types of information are called **modalities**.

A multimodal AI system can use information from multiple modalities together to understand a user's request and generate a relevant response.

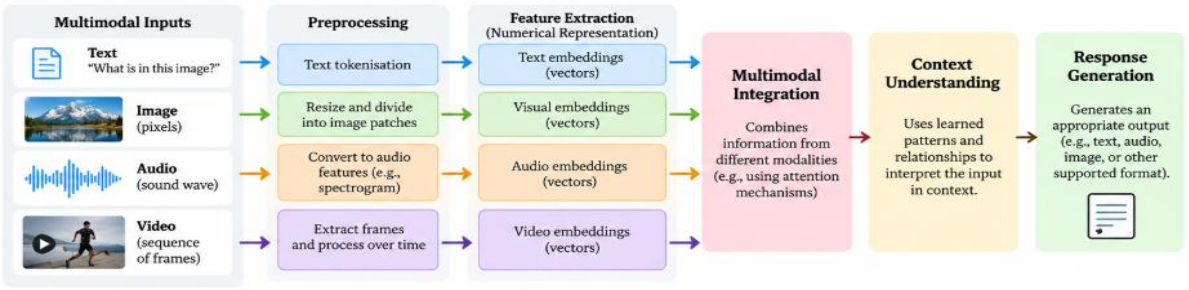
❖ Modalities

A modality refers to a type or format of data that an AI model can process or generate.
Multimodal AI models can work with multiple types of data together.

- Text: Written information, such as questions, instructions, and documents.
- Image: Visual information, such as photographs, charts, and diagrams.
- Audio: Sound information, such as speech, music, and recordings.
- Video: A sequence of visual frames over time, sometimes accompanied by audio.

Example: A multimodal AI model can analyse a picture, understand a question about it, and generate a text response.
Note: The modalities supported depend on the model and its capabilities.

❖ How Multimodal AI Works



The exact internal architecture can be different for different models.

Some systems use separate components to process different modalities before combining their representations. Other systems are designed to handle multiple modalities more directly.
The main idea is that information from different sources can be used together to understand the user's request.

❖ Cross-modal Understanding

Cross-modal understanding means using information from different modalities together.
For example, a model can use:

- An image to understand visual information
- Text to understand the user's question

and combine both to generate an answer.
This allows the AI system to connect different forms of information rather than treating every input type separately.

❖ How Multimodal Models Process Text, Images, Audio, and Video

Multimodal AI models can process different types of data and combine relevant information to generate meaningful responses. The processing method depends on the input type and the model's architecture.

Step 1: Text Processing

The model divides text into smaller units called tokens and converts them into numerical representations called embeddings. These representations help the model process language and understand the user's instructions.
Example: A user asks, "Describe what is happening in this video."

Step 2: Image Processing
The model processes images using a vision encoder. Images divided into smaller regions called patches, which are converted into numerical representations. These representations capture visual features such as objects, shapes, colours, and spatial relationships.
Example: The model identifies a person, a car, and a road in an image.

Step 3: Audio Processing
The model processes audio using techniques designed for sound signals. Depending on the model, audio may be converted into numerical features or processed directly by an audio encoder. These representations can help the model recognise speech, identify sounds, and interpret spoken information.
Example: The model processes a recording and identifies the words spoken by a person.

Step 4: Video Processing
Video contains a sequence of frames displayed over time. The model may extract selected frames and process their visual information to identify objects, actions, and changes between frames. If the model supports audio input, it may also analyse the video's audio track.
Example: The model analyses a video and identifies that a person enters a room, sits on a chair, and starts speaking.

Step 5: Combining Different Modalities
After processing the available inputs, the model connects relevant information using techniques such as attention mechanisms or other multimodal integration methods. This helps it understand relationships between text, images, audio, and video.
Example: A user provides a video with audio and asks, "What is the person doing, and what are they saying?" A capable multimodal model can use the visual information to describe the person's actions and the audio information to identify their spoken words.

Step 6: Response Generation
The model uses the processed information and the user's instructions to generate an appropriate output. Depending on its capabilities, the output may be text, audio, images, or another supported format.
Example: After analysing a video, the model generates a text summary describing the actions and dialogue it could identify.

Note: Not every multimodal model supports all four input types. The exact processing techniques and supported outputs depend on the model's architecture and API.

❖ Multimodal AI and Hallucination
Multimodal AI can also produce incorrect or unsupported information.
For example, a model may:

- Incorrectly identify an object.
- Misinterpret a diagram.s
- Misread text in an image.
- Describe something that is not actually present.
- Make an unsupported statement about unclear visual information.

Therefore, adding more modalities does not automatically guarantee that the AI's response is correct.

❖ Practical Implementation – Multimodal AI
A practical implementation of Multimodal AI is available in the Jupyter Notebook linked below. The notebook demonstrates how a multimodal model processes an image along with a text prompt and generates a response. You can access the project from these link :
Multimodal AI